

2.4 Unit Tangent, Unit Normal, and Binormal Vectors Worksheet

1. Let $\mathbf{r}(t) = \langle t^2, t^2 \sin(t), t^2 \cos(t) \rangle$.

- a) Give parametric equations for the line tangent to $\mathbf{r}(t)$ at $t = \pi$.

When $t = \pi$, we see that $\mathbf{r}(\pi) = \langle \pi^2, 0, -\pi^2 \rangle$ and so $\mathbf{r}(t)$ reaches the point $(\pi^2, 0, -\pi^2)$.

We see that $\mathbf{r}'(t) = \langle 2t, 2t \sin(t) + t^2 \cos(t), 2t \cos(t) - t^2 \sin(t) \rangle$ and thus $\mathbf{r}'(\pi) = \langle 2\pi, -\pi^2, -2\pi \rangle$ is a direction vector for the tangent line.

Therefore the parametric equations for the tangent line are:
$$\begin{cases} x = 2\pi t + \pi^2 \\ y = -\pi^2 t \\ z = -2\pi t - \pi^2 \end{cases}.$$

- b) Determine the unit tangent vector to $\mathbf{r}(t)$ at $t = \pi$.

We know that

$$\mathbf{T}(\pi) = \frac{\mathbf{r}'(\pi)}{\|\mathbf{r}'(\pi)\|} = \frac{\langle 2\pi, -\pi^2, -2\pi \rangle}{\sqrt{4\pi^2 + \pi^4 + 4\pi^2}} = \frac{1}{\pi\sqrt{8 + \pi^2}} \langle 2\pi, -\pi^2, -2\pi \rangle = \left\langle \frac{2}{\sqrt{8 + \pi^2}}, \frac{-\pi}{\sqrt{8 + \pi^2}}, \frac{-2}{\sqrt{8 + \pi^2}} \right\rangle$$

2. Consider the curve defined by $\mathbf{r}(t) = \langle t, 4 - t^2, \ln(t) \rangle$.

- a) Find the point on the curve at which the tangent vector is parallel to the vector $\langle 2, -1, 8 \rangle$.

We see that $\mathbf{r}'(t) = \left\langle 1, -2t, \frac{1}{t} \right\rangle$ and we need $k \left\langle 1, -2t, \frac{1}{t} \right\rangle = \langle 2, -1, 8 \rangle$ for some value of k . Clearly we must have that $k = 2$ otherwise the x -component will not match.

Thus we must have $2 \left\langle 1, -2t, \frac{1}{t} \right\rangle = \langle 2, -1, 8 \rangle$, which implies that $\left\langle 2, -4t, \frac{2}{t} \right\rangle = \langle 2, -1, 8 \rangle$, which then implies that $t = \frac{1}{4}$.

Therefore, the tangent vector is parallel to the vector $\langle 2, -1, 8 \rangle$ when $t = \frac{1}{4}$. This means that the

curve is at $\mathbf{r}\left(\frac{1}{4}\right) = \left\langle \frac{1}{4}, \frac{63}{16}, \ln\left(\frac{1}{4}\right) \right\rangle$ which means that we are at the point $\left(\frac{1}{4}, \frac{63}{16}, \ln\left(\frac{1}{4}\right)\right)$.

- b) Find parametric equations of the tangent line at the point you found in (a).

We can use $\langle 2, -1, 8 \rangle$ as a direction vector and we have $\left(\frac{1}{4}, \frac{63}{16}, \ln\left(\frac{1}{4}\right) \right)$ as a point on the line.

Therefore the parametric equations for the tangent line are:
$$\begin{cases} x = 2t + \frac{1}{4} \\ y = -t + \frac{63}{16} \\ z = 8t + \ln\left(\frac{1}{4}\right) \end{cases}.$$

3. For the curve given by $\mathbf{r}(t) = \langle 4\sin(t), 3t, 4\cos(t) \rangle$.

- a) Find the unit tangent vector $\mathbf{T}(t)$

We note that $\mathbf{r}'(t) = \langle 4\cos(t), 3, -4\sin(t) \rangle$.

Therefore,

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|} = \frac{\langle 4\cos(t), 3, -4\sin(t) \rangle}{\sqrt{16\cos^2(t) + 9 + 16\sin^2(t)}} = \frac{\langle 4\cos(t), 3, -4\sin(t) \rangle}{\sqrt{25}} = \left\langle \frac{4}{5}\cos(t), \frac{3}{5}, -\frac{4}{5}\sin(t) \right\rangle.$$

- b) Find the unit normal vector $\mathbf{N}(t)$.

We know that

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|} = \frac{\left\langle -\frac{4}{5}\sin(t), 0, -\frac{4}{5}\cos(t) \right\rangle}{\sqrt{\frac{16}{25}\sin^2(t) + 0 + \frac{16}{25}\cos^2(t)}} = \frac{\left\langle -\frac{4}{5}\sin(t), 0, -\frac{4}{5}\cos(t) \right\rangle}{\sqrt{\frac{16}{25}}} = \langle -\sin(t), 0, -\cos(t) \rangle$$

c) Find the binormal vector $\mathbf{B}(t)$.

We know that

$$\begin{aligned}\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t) &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{4}{5}\cos(t) & \frac{3}{5} & -\frac{4}{5}\sin(t) \\ -\sin(t) & 0 & -\cos(t) \end{vmatrix} = \left\langle -\frac{3}{5}\cos(t), -\left(-\frac{4}{5}\cos^2(t) - \frac{4}{5}\sin^2(t)\right), \frac{3}{5}\sin(t) \right\rangle \\ &= \left\langle -\frac{3}{5}\cos(t), \frac{4}{5}, \frac{3}{5}\sin(t) \right\rangle\end{aligned}$$

4. Consider the curve $\mathbf{r}(t) = \left\langle t, t^2, \frac{1}{3}t^3 \right\rangle$.

a) Find the unit tangent $\mathbf{T}$ at the point $(0, 0, 0)$.

First we see that $\mathbf{r}(t)$ reaches $(0, 0, 0)$ when $t = 0$. We note that $\mathbf{r}'(t) = \langle 1, 2t, t^2 \rangle$.

$$\text{Therefore, } \mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|} = \frac{\langle 1, 2t, t^2 \rangle}{\sqrt{1+4t^2+t^4}} = \left\langle \frac{1}{\sqrt{1+4t^2+t^4}}, \frac{2t}{\sqrt{1+4t^2+t^4}}, \frac{t^2}{\sqrt{1+4t^2+t^4}} \right\rangle.$$

$$\text{So when } t = 0 \text{ we obtain: } \mathbf{T}(0) = \frac{\mathbf{r}'(0)}{\|\mathbf{r}'(0)\|} = \frac{\langle 1, 0, 0 \rangle}{\sqrt{1+0+0}} = \langle 1, 0, 0 \rangle.$$

b) Find the binormal $\mathbf{B}$ at the point $(0,0,0)$.

We know that $\mathbf{B}(t) = \frac{\mathbf{r}'(t) \times \mathbf{r}''(t)}{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}$ and that $\mathbf{r}(t)$ reaches $(0,0,0)$ when $t=0$. Therefore, we wish

to find $\mathbf{B}(0) = \frac{\mathbf{r}'(0) \times \mathbf{r}''(0)}{\|\mathbf{r}'(0) \times \mathbf{r}''(0)\|}$.

We calculate:

$$\mathbf{r}'(t) = \langle 1, 2t, t^2 \rangle \rightarrow \mathbf{r}'(0) = \langle 1, 0, 0 \rangle$$

$$\mathbf{r}''(t) = \langle 0, 2, 2t \rangle \rightarrow \mathbf{r}''(0) = \langle 0, 2, 0 \rangle$$

$$\mathbf{r}'(0) \times \mathbf{r}''(0) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 0 \\ 0 & 2 & 0 \end{vmatrix} = \langle 0, 0, 2 \rangle$$

$$\|\mathbf{r}'(0) \times \mathbf{r}''(0)\| = \sqrt{0+0+4} = 2$$

$$\text{Therefore, } \mathbf{B}(0) = \frac{\mathbf{r}'(0) \times \mathbf{r}''(0)}{\|\mathbf{r}'(0) \times \mathbf{r}''(0)\|} = \frac{\langle 0, 0, 2 \rangle}{2} = \langle 0, 0, 1 \rangle$$

c) Determine the equation of the plane which contains $\mathbf{T}$ and $\mathbf{N}$.

To create the equation of a plane we need a point in the plane and a normal vector to the plane.

$(0,0,0)$ is a point in the plane and $\mathbf{B}(0)$ serves as a normal vector.

Therefore, the equation of the plane is:

$$0(x-0) + 0(y-0) + 1(z-0) = 0 \rightarrow z = 0$$

Notice that this means that the xy-plane is the plane which is tangent to the curve at the point $(0,0,0)$!